fn sample_truncated_negative_binomial_rational

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This document proves that the implementation of `sample_truncated_negative_binomial_rational` in `mod.rs` satisfies its proof definition.

1 Hoare Triple

Precondition

`eta` $\in \mathbb{Q}_{>0}$ and `x` $\in \mathbb{Q}_{>0}$.

Pseudocode

```
1
2 def sample_truncated_negative_binomial_rational(eta, x) -> int:
3     while True:
4         k = sample_negative_binomial_rational(eta, x)  #
5         if k != 0:  #
6             return k
```

Postcondition

For any setting of the input parameters `eta` and `x` such that the given preconditions hold, `sample_truncated_negative_binom` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as a negative binomial random variable with shape `eta` and success probability $1 - e^{-x}$, parameterized as the number of failures before the `eta`-th success, conditioned on `out > 0`.

2 Proof

Assume the preconditions are met. Let K denote a random variable distributed as the output of `sample_negative_binomial_` `x)`. By its proof definition, for every $k \in \mathbb{N}_0$,

$$\Pr[K = k] = (1 - e^{-x})^{\eta} e^{-xk} \prod_{i=0}^{k-1} \frac{\eta + i}{i + 1}.$$

Lemma 2.1. `sample_truncated_negative_binomial_rational` only returns `Err(e)` when there is a lack of system entropy.

Proof. The only fallible call is to `sample_negative_binomial_rational` on line 4. By its proof definition, any returned error must be due to a lack of system entropy. □

Lemma 2.2. If the outcome of `sample_truncated_negative_binomial_rational` is `Ok(out)`, then `out` is distributed as the law of K conditioned on $K > 0$.

Proof. In each loop iteration, line 4 draws an independent sample distributed as K. The function returns the first such draw satisfying the predicate on line 5. Since

$$\Pr[K > 0] = 1 - \Pr[K = 0] = 1 - (1 - e^{-x})^{\eta} > 0,$$

the loop terminates almost surely. For any $k \in \mathbb{N}_{>0}$,

$$\Pr[\texttt{out} = k] = \sum_{t=1}^{\infty} \Pr[K = 0]^{t-1} \Pr[K = k] = \frac{\Pr[K = k]}{1 - \Pr[K = 0]}.$$

Therefore the output distribution is exactly the conditional distribution of K given $K > 0$. □

Lemma 2.3. If $\gamma := 1 - e^{-x}$, then the output distribution equals

$$\Pr[\texttt{out} = k] = \frac{(1-\gamma)^k}{\gamma^{-\eta} - 1} \prod_{i=0}^{k-1} \frac{\eta + i}{i + 1} \qquad \text{for } k \in \mathbb{N}_{>0}.$$

Proof. For any $k \in \mathbb{N}_{>0}$,

$$\begin{aligned}
\Pr[\texttt{out} = k] &= \frac{(1 - e^{-x})^{\eta} e^{-xk} \prod_{i=0}^{k-1} \frac{\eta+i}{i+1}}{1 - (1 - e^{-x})^{\eta}} \\
&= \frac{\gamma^{\eta} (1-\gamma)^k \prod_{i=0}^{k-1} \frac{\eta+i}{i+1}}{1 - \gamma^{\eta}} \\
&= \frac{(1-\gamma)^k}{\gamma^{-\eta} - 1} \prod_{i=0}^{k-1} \frac{\eta + i}{i + 1}.
\end{aligned}$$

This is exactly the claimed pmf. □

Proof. 1 holds by Lemma 2.1 and Lemma 2.2. Lemma 2.3 identifies the conditional law with the truncated negative binomial family. □